JI ZHOU

Robotics Engineering | Humanoid Robot Learning and Deployment

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Project Portfolio

RESEARCH PROFILE

Master's student specialising in humanoid locomotion, imitation learning and sim-to-real deployment. Research interests: whole-body motion, loco-manipulation and end-to-end robot development.

EDUCATION

Chongqing University | M.Eng. in Robotics Engineering 2025 - Present

GPA: **3.66/4.00** | *Research focus: humanoid robotics and embodied intelligence.*

Chongqing University | B.Eng. in Mechatronics Engineering 2021 - 2025

GPA: **3.06/4.00**

RESEARCH AND ENGINEERING EXPERIENCE

Full-Size Humanoid Robot Learning and Sim-to-Real Deployment

Primary contributor: policy training, deployment and system integration Mar 2026 - Present

- Built a locomotion pipeline in MJLab using PPO and an asymmetric actor-critic architecture; applied velocity and terrain curricula to improve policy generalisation.
- Applied domain randomisation over robot mass, control latency, joint friction, foot-contact stiffness and encoder bias to improve robustness and sim-to-real transfer.
- Developed a ROS2 framework for policy inference, command delivery, action execution, observation feedback and logging; deployed policies at **50 Hz** using C++ and ONNX Runtime.
- Implemented Ethernet-to-CAN FD motor communication via UDP, Cannelloni and SocketCAN for DAMIAO actuators through DM-LinkX.
- Developed parallel-ankle kinematics for virtual-to-motor position, velocity and torque mapping; built motor, IMU, communication and tracking diagnostics. Completed physical integration and debugging, achieving **stable real-robot walking**.

Motion Training and Fleet Control for Eight Unitree G1 Robots

Motion training, fleet-control development and on-site deployment Sep 2025 - Feb 2026

- Performed dance motion retargeting and policy training for Unitree G1 humanoid robots using GMR and BeyondMimic.
- Designed a UDP fleet-control system for **eight robots**, supporting fleet, group and individual operation through a graphical operator interface.
- Integrated stage entrance and exit sequencing, synchronised dance triggering, run-time policy switching and robot status monitoring.
- Used timestamp-based synchronisation and fault-tolerant execution to handle network fluctuations and temporary disconnections; deployed the complete system in a **live eight-robot stage performance**.

UNDERGRADUATE GRADUATION PROJECT

LQR-Based Design and Control of a Biped Wheeled-Legged Robot 2025

- Designed a biped wheeled-legged robot and implemented linear quadratic regulator (LQR) control on an STM32 microcontroller.
- Realised locomotion, body-height adjustment and jumping on the physical robot, combining mechanical design with embedded control implementation.

SELECTED ROBOTICS COMPETITION PROJECTS

Agricultural Scallion Transplanter 2024

National Second Prize, National College Mechanical Innovation Competition

- Designed the vehicle electrical and control architecture, covering chassis drive, actuator control, sensor integration and remote operation.
- Selected motors, sensors and structural components to support the mechanical and electrical system design.
- Developed microcontroller firmware for chassis drive and motor control; implemented serial and CAN communication and remote-control functions.

Sorting Robot | Team Leader 2023

National Second Prize, China Robot Competition and RoboCup China Open

- Led mechanical design and fabrication of the sorting robot, including its body, manipulator, mobile base and storage mechanism.
- Selected motors and sensors; developed STM32F407-based drive software, vision-interface functions and robot control logic.
- Implemented serial and I2C communication for embedded device interfacing; used SolidWorks for mechanical design and Keil for embedded development.

TECHNICAL SKILLS

Programming C++, Python, embedded C

Robot learning PPO, asymmetric actor-critic, imitation learning, motion retargeting, curriculum learning and domain randomisation

Robotics software ROS2, MJLab, GMR, BeyondMimic, ONNX Runtime

Embedded systems and control STM32, LQR, actuator and sensor integration, motor and IMU diagnostics, real-time control

Communication UDP, CAN, CAN FD, SocketCAN, Cannelloni, serial communication, I2C

Mechanical design Robot mechanism design, parallel-ankle kinematics, SolidWorks, fabrication and system integration

HONORS AND AWARDS

Merit Student, Chongqing University 2026

Chongqing University Scholarship 2026

National Second Prize, National College Mechanical Innovation Competition 2024

National Second Prize, China Robot Competition and RoboCup China Open 2023